

HW5

Regression Analysis

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Table of contents

Question 9	2
Load package and data	2
EDA	2
(a)	3
(b)	4
(c)	4
(e)	5
(f)	7
Question 10	10
Load dataset and EDA	10
(a)	12
(b)	13
Intercept ($\beta_0 = 5.6606$):	13
Quantitative Variables:	13
Categorical Variables:	14
Model Fit:	14
(c)	14
Substituting the coefficients:	14
Handling qualitative variables:	15
Model Interpretation:	15
(d)	15
Coefficients Table	15
Decision Rule	15
Results	16
Conclusion	16
(e)	16
Significant Predictors	16
R Code for Smaller Model	16
(f)	17
Comparing Model Fits: lm.reg1 (Model a) and lm.smaller (Model e)	17
Comparison Table	17
Interpretation	18
Conclusion	18
Question 3	18
Load dataset(Credit.csv) and EDA	18
M1	20

M3	20
M4	21
M5	21
Model Summaries:	22
Conclusion:	22

Question 9

Load package and data

```
library(dplyr)
library(ggplot2)
library(GGally)
# install.packages("GGally")
auto <- read.csv("Auto.csv")
head(auto)
```

```
X mpg cylinders displacement horsepower weight acceleration year origin
1 18 8 307 130 3504 12.0 70 1
2 15 8 350 165 3693 11.5 70 1
3 18 8 318 150 3436 11.0 70 1
4 16 8 304 150 3433 12.0 70 1
5 17 8 302 140 3449 10.5 70 1
6 15 8 429 198 4341 10.0 70 1

name
1 chevrolet chevelle malibu
2 buick skylark 320
3 plymouth satellite
4 amc rebel sst
5 ford torino
6 ford galaxie 500
```

EDA

```
library(Hmisc)
latex(describe(auto), file = "", caption.placement = "top")
```

10 Variables										auto 392 Observations			
X													
n 392	missing 0	distinct 392	Info 1	Mean 198.5	Gmd 132.3	.05 20.55	.10 41.10	.25 99.75	.50 198.50	.75 296.25	.90 357.90	.95 377.45	
lowest: 1 2 3 4 5, highest: 393 394 395 396 397													
mpg													
n 392	missing 0	distinct 127	Info 0.999	Mean 23.45	Gmd 8.879	.05 13.00	.10 14.00	.25 17.00	.50 22.75	.75 29.00	.90 34.19	.95 37.00	
lowest: 9 10 11 12 13, highest: 43.4 44 44.3 44.6 46.6													

cylinders

n	missing	distinct	Info	Mean	Gmd
392	0	5	0.842	5.472	1.798

Value	3	4	5	6	8
Frequency	199	83	103		
Proportion	0.010	0.508	0.008	0.212	0.263

For the frequency table, variable is rounded to the nearest 0

displacement

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
392	0	81	0.999	194.4	115.7	85.0	90.0	105.0	151.0	275.8	350.0	400.0

lowest: 68 70 71 72 76, highest: 400 429 440 454 455

horsepower

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
392	0	93	0.999	104.5	41.48	60.55	67.00	75.00	93.50	126.00	157.70	180.00

lowest: 46 48 49 52 53, highest: 210 215 220 225 230

weight

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
392	0	346	1	2978	964.1	1932	1990	2225	2804	3615	4278	4464

lowest: 1613 1649 1755 1760 1773, highest: 4951 4952 4955 4997 5140

acceleration

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
392	0	95	0.999	15.54	3.079	11.26	12.00	13.78	15.50	17.02	19.00	20.23

lowest: 8 8.5 9 9.5 10 , highest: 22.2 23.5 23.7 24.6 24.8

year

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
392	0	13	0.994	75.98	4.245	70	71	73	76	79	81	82

Value	70	71	72	73	74	75	76	77	78	79	80	81	82
Frequency	29	27	28	40	26	30	34	28	36	29	27	28	30
Proportion	0.074	0.069	0.071	0.102	0.066	0.077	0.087	0.071	0.092	0.074	0.069	0.071	0.077

For the frequency table, variable is rounded to the nearest 0

origin

n	missing	distinct	Info	Mean	Gmd
392	0	3	0.742	1.577	0.7926

Value	1	2	3
Frequency	245	68	79
Proportion	0.625	0.173	0.202

For the frequency table, variable is rounded to the nearest 0

name

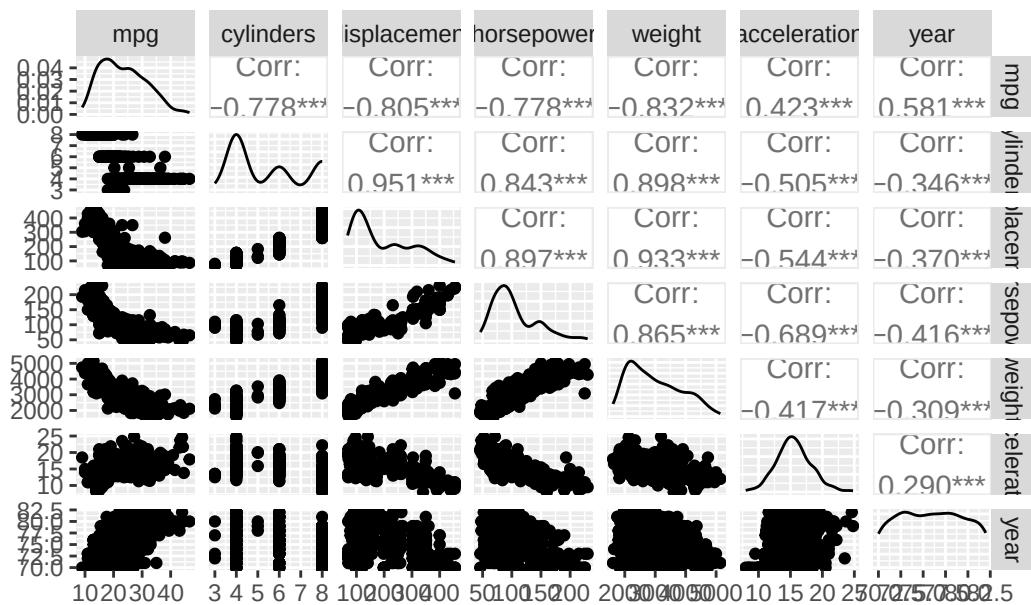
n	missing	distinct
392	0	301

lowest: amc ambassador broughton amc ambassador dpl amc ambassador sst amc concord amc concord d/l
highest: vw dasher (diesel) vw pickup vw rabbit vw rabbit c (diesel) vw rabbit custom

(a)

```
# a.  
Scatter_Matrix <- ggpairs(auto, columns = c(2:8),  
                           title = "Scatter Plot Matrix for Auto Dataset",  
                           axisLabels = "show")  
# ggsave("Scatter plot matrix.png", Scatter_Matrix, width = 7, height = 7, units = "in")  
Scatter_Matrix
```

Scatter Plot Matrix for Auto Dataset



(b)

```
library(knitr)
cor_matrix <- cor(auto[-c(1, 9, 10)])
kable(cor_matrix)
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	year
mpg	1.0000000	-	-0.8051269	-	-	0.4233285	0.5805410
cylinders		1.0000000	0.9508233	0.7784268	0.8322442	-0.5046834	-
displacement			1.0000000	0.8972570	0.9329944	-0.5438005	0.3456474
horsepower				1.0000000	0.8645377	-0.6891955	0.3698552
weight					1.0000000	-0.4168392	0.4163615
acceleration						1.0000000	0.3091199
year							1.0000000

(c)

因為origin應為nominal variable，因此將其作成dummy variable。

```
## data without the column name
auto_no_name <- auto[-c(1, 10)]
## create dummy variables
```

```
class(auto_no_name$origin)
```

```
[1] "integer"
```

```
auto_no_name$origin <- as.factor(auto_no_name$origin)
class(auto_no_name$origin)
```

```
[1] "factor"
```

```
contrasts(auto_no_name$origin) <- contr.treatment(3, base = 1)
```

```
regression <- lm(mpg ~ ., data = auto_no_name)
summary(regression)
```

Call:

```
lm(formula = mpg ~ ., data = auto_no_name)
```

Residuals:

```
    Min     1Q   Median     3Q      Max
-9.0095 -2.0785 -0.0982  1.9856 13.3608
```

Coefficients:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.795e+01  4.677e+00  -3.839 0.000145 ***
cylinders    -4.897e-01  3.212e-01  -1.524 0.128215
displacement  2.398e-02  7.653e-03   3.133 0.001863 **
horsepower   -1.818e-02  1.371e-02  -1.326 0.185488
weight       -6.710e-03  6.551e-04 -10.243 < 2e-16 ***
acceleration  7.910e-02  9.822e-02   0.805 0.421101
year          7.770e-01  5.178e-02  15.005 < 2e-16 ***
origin2       2.630e+00  5.664e-01   4.643 4.72e-06 ***
origin3       2.853e+00  5.527e-01   5.162 3.93e-07 ***
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.307 on 383 degrees of freedom

Multiple R-squared: 0.8242, Adjusted R-squared: 0.8205

F-statistic: 224.5 on 8 and 383 DF, p-value: < 2.2e-16

- 從上方報表呈現之結果，可知在 null hypothesis為全部 $\beta = 0$ 時，其拒絕其虛無假設，因此 predictors 和 response 之間有關係。
- **displacement, weight and year** have a statistically significant relationship to the response.
- 0.777 is the coefficient for the **year** variable suggest.

(e)

```
interaction_model <- lm(mpg ~ (.)^2, data = auto_no_name)
summary(interaction_model)
```

Call:

```
lm(formula = mpg ~ (.)^2, data = auto_no_name)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-7.6008	-1.2863	0.0813	1.2082	12.0382

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.401e+01	5.147e+01	0.855	0.393048
cylinders	3.302e+00	8.187e+00	0.403	0.686976
displacement	-3.529e-01	1.974e-01	-1.788	0.074638 .
horsepower	5.312e-01	3.390e-01	1.567	0.117970
weight	-3.259e-03	1.820e-02	-0.179	0.857980
acceleration	-6.048e+00	2.147e+00	-2.818	0.005109 **
year	4.833e-01	5.923e-01	0.816	0.415119
origin2	-3.517e+01	1.260e+01	-2.790	0.005547 **
origin3	-3.765e+01	1.426e+01	-2.640	0.008661 **
cylinders:displacement	-6.316e-03	7.106e-03	-0.889	0.374707
cylinders:horsepower	1.452e-02	2.457e-02	0.591	0.555109
cylinders:weight	5.703e-04	9.044e-04	0.631	0.528709
cylinders:acceleration	3.658e-01	1.671e-01	2.189	0.029261 *
cylinders:year	-1.447e-01	9.652e-02	-1.499	0.134846
cylinders:origin2	-7.210e-01	1.088e+00	-0.662	0.508100
cylinders:origin3	1.226e+00	1.007e+00	1.217	0.224379
displacement:horsepower	-5.407e-05	2.861e-04	-0.189	0.850212
displacement:weight	2.659e-05	1.455e-05	1.828	0.068435 .
displacement:acceleration	-2.547e-03	3.356e-03	-0.759	0.448415
displacement:year	4.547e-03	2.446e-03	1.859	0.063842 .
displacement:origin2	-3.364e-02	4.220e-02	-0.797	0.425902
displacement:origin3	5.375e-02	4.145e-02	1.297	0.195527
horsepower:weight	-3.407e-05	2.955e-05	-1.153	0.249743
horsepower:acceleration	-3.445e-03	3.937e-03	-0.875	0.382122
horsepower:year	-6.427e-03	3.891e-03	-1.652	0.099487 .
horsepower:origin2	-4.869e-03	5.061e-02	-0.096	0.923408
horsepower:origin3	2.289e-02	6.252e-02	0.366	0.714533
weight:acceleration	-6.851e-05	2.385e-04	-0.287	0.774061
weight:year	-8.065e-05	2.184e-04	-0.369	0.712223
weight:origin2	2.277e-03	2.685e-03	0.848	0.397037
weight:origin3	-4.498e-03	3.481e-03	-1.292	0.197101
acceleration:year	6.141e-02	2.547e-02	2.412	0.016390 *
acceleration:origin2	9.234e-01	2.641e-01	3.496	0.000531 ***
acceleration:origin3	7.159e-01	3.258e-01	2.198	0.028614 *
year:origin2	2.932e-01	1.444e-01	2.031	0.043005 *
year:origin3	3.139e-01	1.483e-01	2.116	0.035034 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.628 on 356 degrees of freedom

Multiple R-squared: 0.8967, Adjusted R-squared: 0.8866

F-statistic: 88.34 on 35 and 356 DF, p-value: < 2.2e-16

檢查統計顯著的交互作用

```
significant_interactions <- summary(interaction_model)$coefficients
```

```
significant_interactions <- significant_interactions[which(significant_interactions[,4] < 0.05),]
```

```
print("統計顯著的交互作用項:")
```

```
[1] "統計顯著的交互作用項:"
```

```
print(significant_interactions)
```

	Estimate	Std. Error	t value	Pr(> t)
acceleration	-6.04827289	2.14663070	-2.817566	0.0051089710
origin2	-35.16513846	12.60196356	-2.790449	0.0055469796
origin3	-37.64639178	14.26129623	-2.639759	0.0086614400
cylinders:acceleration	0.36581508	0.16713092	2.188794	0.0292609322
acceleration:year	0.06141264	0.02546586	2.411568	0.0163896470
acceleration:origin2	0.92339618	0.26409862	3.496407	0.0005311998
acceleration:origin3	0.71592928	0.32576288	2.197701	0.0286139299
year:origin2	0.29322251	0.14437874	2.030926	0.0430051989
year:origin3	0.31388902	0.14833731	2.116049	0.0350343723

(f)

```
# 原始變數模型
original_model <- lm(mpg ~ ., data = auto_no_name)
summary(original_model)
```

Call:

```
lm(formula = mpg ~ ., data = auto_no_name)
```

Residuals:

Min	1Q	Median	3Q	Max
-9.0095	-2.0785	-0.0982	1.9856	13.3608

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.795e+01	4.677e+00	-3.839	0.000145 ***
cylinders	-4.897e-01	3.212e-01	-1.524	0.128215
displacement	2.398e-02	7.653e-03	3.133	0.001863 **
horsepower	-1.818e-02	1.371e-02	-1.326	0.185488
weight	-6.710e-03	6.551e-04	-10.243	< 2e-16 ***
acceleration	7.910e-02	9.822e-02	0.805	0.421101
year	7.770e-01	5.178e-02	15.005	< 2e-16 ***
origin2	2.630e+00	5.664e-01	4.643	4.72e-06 ***
origin3	2.853e+00	5.527e-01	5.162	3.93e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.307 on 383 degrees of freedom

Multiple R-squared: 0.8242, Adjusted R-squared: 0.8205

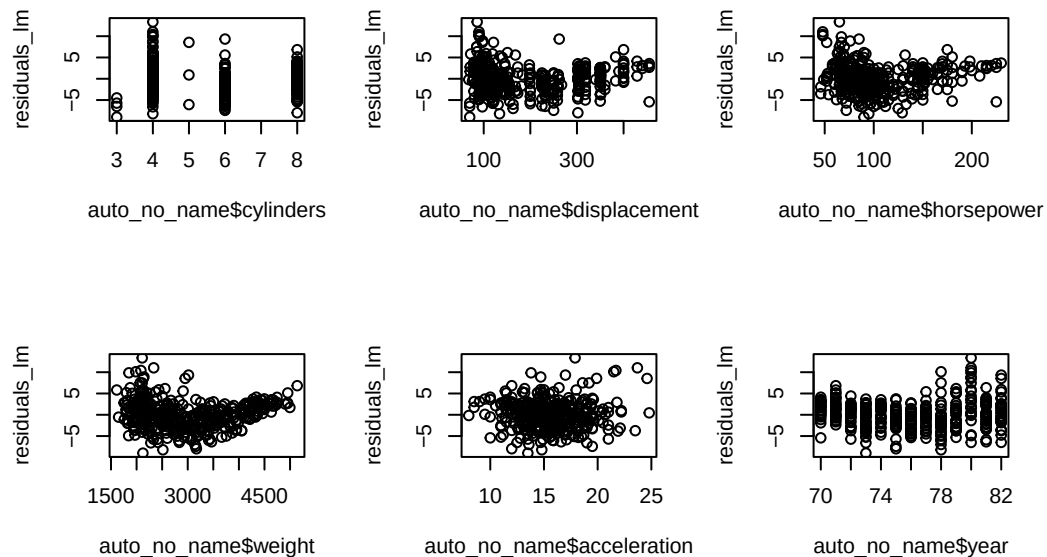
F-statistic: 224.5 on 8 and 383 DF, p-value: < 2.2e-16

```
residuals_lm <- original_model$residuals
names(auto_no_name)
```

```
[1] "mpg"      "cylinders" "displacement" "horsepower" "weight"
[6] "acceleration" "year"      "origin"
```

```
par(mfrow=c(2,3))
plot(x = auto_no_name$cylinders, residuals_lm)
plot(x = auto_no_name$displacement, residuals_lm)
```

```
plot(x = auto_no_name$horsepower, residuals_lm)
plot(x = auto_no_name$weight, residuals_lm)
plot(x = auto_no_name$acceleration, residuals_lm)
plot(x = auto_no_name$year, residuals_lm)
```



包含轉換變數的模型

```
transformed_model1 <- lm(mpg ~ cylinders + I(displacement^2) +
  sqrt(horsepower) + log(weight) + acceleration +
  year + origin, data = auto_no_name)
summary(transformed_model1)
```

Call:

```
lm(formula = mpg ~ cylinders + I(displacement^2) + sqrt(horsepower) +
  log(weight) + acceleration + year + origin, data = auto_no_name)
```

Residuals:

```
Min    1Q  Median    3Q   Max
-9.0565 -1.7563  0.0453  1.6518 12.6699
```

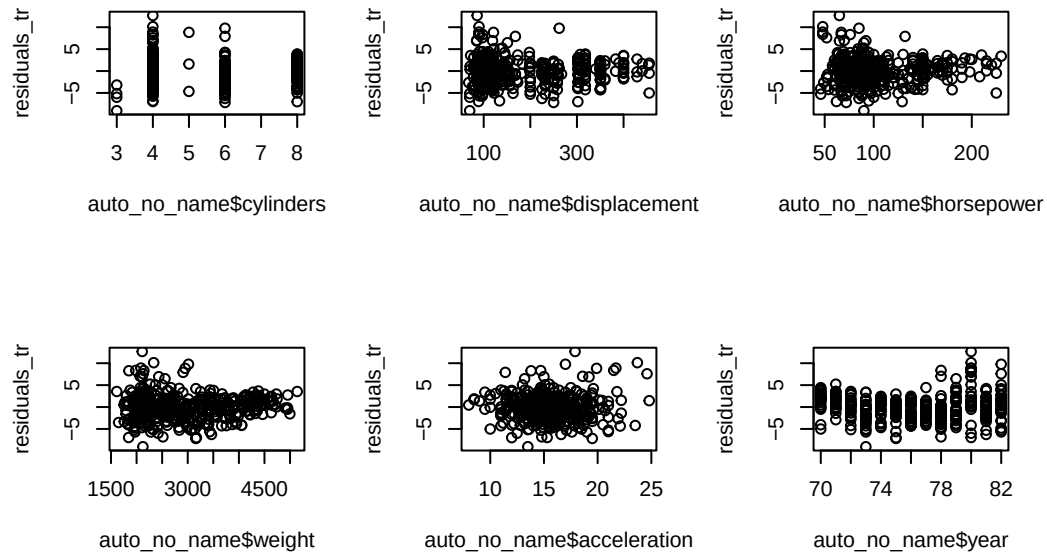
Coefficients:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.216e+02  1.040e+01  11.690 < 2e-16 ***
cylinders    -3.908e-01  2.476e-01  -1.578 0.115342
I(displacement^2) 5.264e-05  9.212e-06  5.713 2.23e-08 ***
sqrt(horsepower) -9.790e-01  2.946e-01  -3.323 0.000975 ***
log(weight)   -1.888e+01  1.659e+00 -11.380 < 2e-16 ***
acceleration   4.427e-03  9.349e-02   0.047 0.962259
year          7.991e-01  4.706e-02  16.980 < 2e-16 ***
origin2        1.952e+00  4.841e-01   4.032 6.68e-05 ***
origin3        1.863e+00  4.872e-01   3.823 0.000154 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

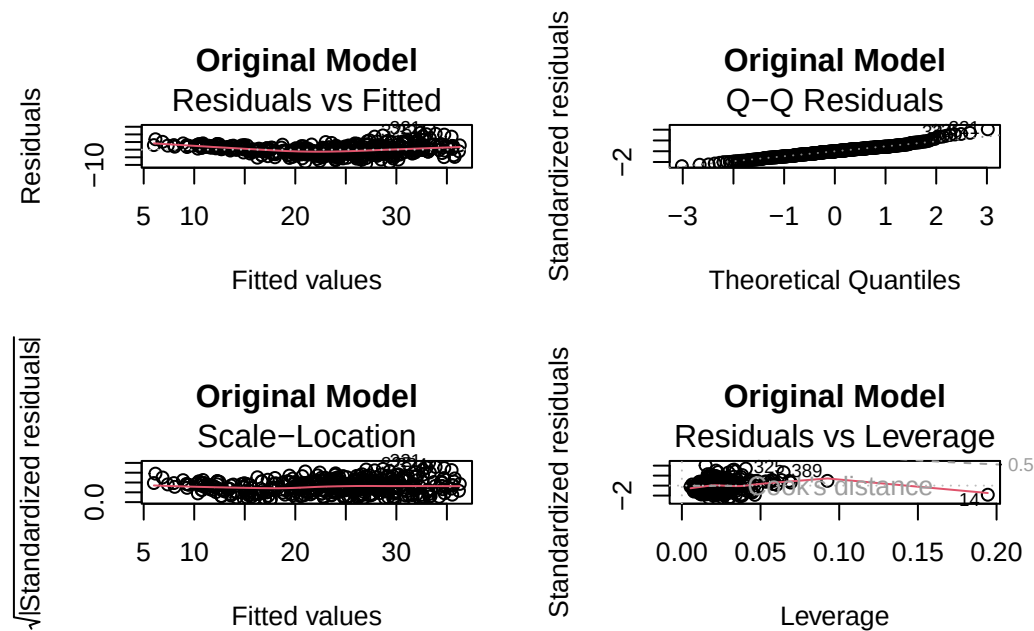
Residual standard error: 2.985 on 383 degrees of freedom

Multiple R-squared: 0.8567, Adjusted R-squared: 0.8537
F-statistic: 286.2 on 8 and 383 DF, p-value: < 2.2e-16

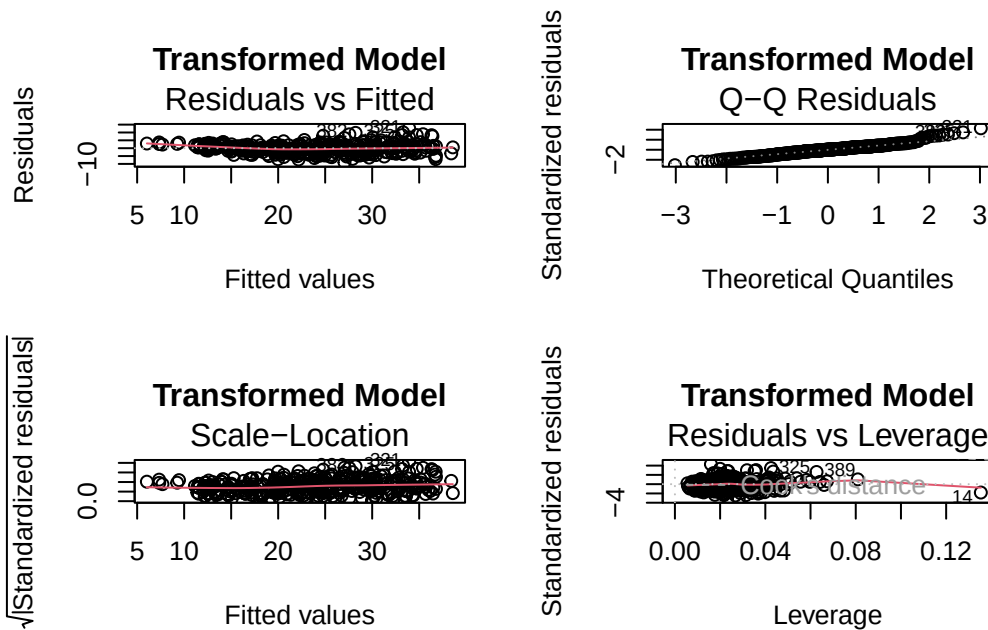
```
residuals_tr <- transformed_model1$residuals
par(mfrow=c(2,3))
plot(x = auto_no_name$cylinders, residuals_tr)
plot(x = auto_no_name$displacement, residuals_tr)
plot(x = auto_no_name$horsepower, residuals_tr)
plot(x = auto_no_name$weight, residuals_tr)
plot(x = auto_no_name$acceleration, residuals_tr)
plot(x = auto_no_name$year, residuals_tr)
```



```
# 視覺化轉換效果
par(mfrow = c(2, 2))
plot(original_model, main = "Original Model")
```



```
plot(transformed_model1, main = "Transformed Model")
```



The residual plots perform better in transformed model if you compare to the original model. Besides, there are some coefficient for some variables is more easier to have a statistically significant.

Question 10

Load dataset and EDA

```
carseat <- read.csv("Carseats.csv")
head(carseat)
```

```
  X Sales CompPrice Income Advertising Population Price ShelfLoc Age Education
1 1 9.50   138    73      11      276 120    Bad 42    17
2 2 11.22   111    48      16      260 83    Good 65    10
3 3 10.06   113    35      10      269 80    Medium 59    12
4 4 7.40    117   100       4      466 97    Medium 55    14
5 5 4.15    141    64       3      340 128    Bad 38    13
6 6 10.81   124   113      13      501 72    Bad 78    16
  Urban US
1 Yes Yes
2 Yes Yes
3 Yes Yes
4 Yes Yes
5 Yes No
6 No Yes
```

```
str(carseat)
```

```
'data.frame': 400 obs. of 12 variables:
 $ X      : int 1 2 3 4 5 6 7 8 9 10 ...
 $ Sales  : num 9.5 11.22 10.06 7.4 4.15 ...
 $ CompPrice : int 138 111 113 117 141 124 115 136 132 132 ...
```

```

$ Income : int 73 48 35 100 64 113 105 81 110 113 ...
$ Advertising: int 11 16 10 4 3 13 0 15 0 0 ...
$ Population : int 276 260 269 466 340 501 45 425 108 131 ...
$ Price : int 120 83 80 97 128 72 108 120 124 124 ...
$ ShelfLoc : chr "Bad" "Good" "Medium" "Medium" ...
$ Age : int 42 65 59 55 38 78 71 67 76 76 ...
$ Education : int 17 10 12 14 13 16 15 10 10 17 ...
$ Urban : chr "Yes" "Yes" "Yes" "Yes" ...
$ US : chr "Yes" "Yes" "Yes" "Yes" ...

```

```

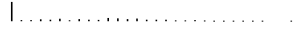
carseat$ShelveLoc <- as.factor(carseat$ShelveLoc)
carseat$Urban <- as.factor(carseat$Urban)
carseat$US <- as.factor(carseat$US)

```

```

library(Hmisc)
latex(describe(carseat), file = "", caption.placement = "top")

```

carseat													
12 Variables 400 Observations													
X													
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95	
400	0	400	1	200.5	133.7	20.95	40.90	100.75	200.50	300.25	360.10	380.05	
lowest: 1 2 3 4 5, highest: 396 397 398 399 400													
Sales													
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95	
400	0	336	1	7.496	3.192	3.149	4.119	5.390	7.490	9.320	11.300	12.442	
lowest: 0 0.16 0.37 0.53 0.91, highest: 13.91 14.37 14.9 15.63 16.27													
CompPrice													
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95	
400	0	73	0.999	125	17.3	.98	106	115	125	135	145	150	
lowest: 77 85 86 88 89, highest: 157 159 161 162 175													
Income													
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95	
400	0	98	1	68.66	32.29	26.00	30.00	42.75	69.00	91.00	107.00	115.00	
lowest: 21 22 23 24 25, highest: 116 117 118 119 120													
Advertising													
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95	
400	0	28	0.952	6.635	7.337	0	0	0	5	12	16	19	
lowest: 0 1 2 3 4, highest: 23 24 25 26 29													
Population													
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95	
400	0	275	1	264.8	170.3	29.0	58.9	139.0	272.0	398.5	467.0	493.1	
lowest: 10 12 13 14 16, highest: 503 504 507 508 509													
Price													
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95	
400	0	101	1	115.8	26.52	.77	.87	100	117	131	146	155	
lowest: 24 49 53 54 55, highest: 166 171 173 185 191													
ShelveLoc													
n	missing	distinct											
400	0	3											
Value	Bad	Good	Medium										
Frequency	96	85	219										
Proportion	0.240	0.212	0.547										

Age

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
400	0	56	1	53.32	18.71	27.00	30.00	39.75	54.50	66.00	76.00	79.00

lowest : 25 26 27 28 29, highest: 76 77 78 79 80

Education

n	missing	distinct	Info	Mean	Gmd
400	0	9	0.987	13.9	3.009

Value	10	11	12	13	14	15	16	17	18
Frequency	48	48	49	43	40	36	47	49	40
Proportion	0.120	0.120	0.122	0.108	0.100	0.090	0.117	0.122	0.100

For the frequency table, variable is rounded to the nearest 0

Urban

n	missing	distinct
400	0	2

Value	No	Yes
Frequency	118	282
Proportion	0.295	0.705

US

n	missing	distinct
400	0	2

Value	No	Yes
Frequency	142	258
Proportion	0.355	0.645

(a)

```
lm.reg1 <- lm(Sales ~ Price + Urban + US, data = carseat)
summary(lm.reg1)
```

Call:

```
lm(formula = Sales ~ Price + Urban + US, data = carseat)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.9206	-1.6220	-0.0564	1.5786	7.0581

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.043469	0.651012	20.036	< 2e-16 ***
Price	-0.054459	0.005242	-10.389	< 2e-16 ***
UrbanYes	-0.021916	0.271650	-0.081	0.936
USYes	1.200573	0.259042	4.635	4.86e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.472 on 396 degrees of freedom

Multiple R-squared: 0.2393, Adjusted R-squared: 0.2335

F-statistic: 41.52 on 3 and 396 DF, p-value: < 2.2e-16

This mean that :

$$Sales = 13.043469 - 0.054459 \times Price + 1.200573 \times US$$

where $US = 0$ when the store is **not** in the US and $US = 1$ when the store is in the US.

(b)

```
lm.all <- lm(Sales ~ (. - X), data = carseat)
summary(lm.all)
```

Call:

```
lm(formula = Sales ~ (. - X), data = carseat)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.8692	-0.6908	0.0211	0.6636	3.4115

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.6606231	0.6034487	9.380	< 2e-16 ***
CompPrice	0.0928153	0.0041477	22.378	< 2e-16 ***
Income	0.0158028	0.0018451	8.565	2.58e-16 ***
Advertising	0.1230951	0.0111237	11.066	< 2e-16 ***
Population	0.0002079	0.0003705	0.561	0.575
Price	-0.0953579	0.0026711	-35.700	< 2e-16 ***
ShelveLocGood	4.8501827	0.1531100	31.678	< 2e-16 ***
ShelveLocMedium	1.9567148	0.1261056	15.516	< 2e-16 ***
Age	-0.0460452	0.0031817	-14.472	< 2e-16 ***
Education	-0.0211018	0.0197205	-1.070	0.285
UrbanYes	0.1228864	0.1129761	1.088	0.277
USYes	-0.1840928	0.1498423	-1.229	0.220

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.019 on 388 degrees of freedom

Multiple R-squared: 0.8734, Adjusted R-squared: 0.8698

F-statistic: 243.4 on 11 and 388 DF, p-value: < 2.2e-16

Below is an interpretation of each coefficient:

Intercept ($\beta_0 = 5.6606$):

The intercept represents the predicted value of the dependent variable (e.g., sales) when all independent variables are at their reference levels or equal to zero. For categorical variables, the reference categories are likely: - *ShelveLoc* = *Bad* - *Urban* = *No* - *US* = *No*

Quantitative Variables:

1. **CompPrice** ($\beta_1 = 0.0928$): For every unit increase in the competitor's price, the dependent variable is expected to increase by 0.0928 units, holding all other variables constant.
2. **Income** ($\beta_2 = 0.0158$): For every unit increase in income, the dependent variable increases by 0.0158 units, holding all other variables constant.
3. **Advertising** ($\beta_3 = 0.1231$): For every unit increase in advertising expenditure, the dependent variable increases by 0.1231 units, holding all other variables constant.
4. **Population** ($\beta_4 = 0.0002$): Population does not appear to have a statistically significant effect ($p = 0.575$). However, the coefficient suggests that for every unit increase in population, the dependent variable increases by 0.0002 units.

5. **Price** ($\beta_5 = -0.0954$): For every unit increase in the price of the product, the dependent variable decreases by 0.0954 units, holding all other variables constant. This negative coefficient indicates that higher prices reduce sales (or whatever the dependent variable represents).
6. **Age** ($\beta_8 = -0.0460$): For every additional year of age, the dependent variable decreases by 0.0460 units, holding all other variables constant.
7. **Education** ($\beta_9 = -0.0211$): Education **does not have a statistically significant effect** ($p = 0.285$). The coefficient suggests that for every unit increase in education, the dependent variable decreases by 0.0211 units.

Categorical Variables:

Categorical variables have been encoded with dummy variables, comparing levels against a reference level.

1. **ShelveLoc:**
 - **Good** ($\beta_6 = 4.8502$): Products on "Good" shelves result in a predicted increase of 4.8502 units in the dependent variable compared to "Bad" shelves, holding all other variables constant.
 - **Medium** ($\beta_7 = 1.9567$): Products on "Medium" shelves result in a predicted increase of 1.9567 units compared to "Bad" shelves.
2. **Urban:**
 - **Yes** ($\beta_{10} = 0.1229$): Urban areas **do not have a statistically significant effect** ($p = 0.277$). The coefficient suggests that products in urban areas might lead to an increase of 0.1229 units compared to non-urban areas.
3. **US:**
 - **Yes** ($\beta_{11} = -0.1841$): The US variable is **not statistically significant** ($p = 0.220$). However, the coefficient suggests that products in the US lead to a decrease of 0.1841 units compared to non-US locations.

Model Fit:

- **Residual Standard Error (RSE):** 1.019 indicates the typical deviation of observed values from the regression line.
- $R^2 = 0.8734$: The model explains 87.34% of the variability in the dependent variable.
- **Adjusted $R^2 = 0.8698$:** Adjusted for the number of predictors, it still reflects a strong fit.
- **F-statistic** ($p < 2.2e^{-16}$): The overall model is highly statistically significant.

(c)

The multiple linear regression model can be written as follows:

$$Y = \beta_0 + \beta_1 \cdot \text{CompPrice} + \beta_2 \cdot \text{Income} + \beta_3 \cdot \text{Advertising} + \beta_4 \cdot \text{Population} + \beta_5 \cdot \text{Price} + \beta_6 \cdot \text{ShelveLocGood} + \beta_7 \cdot \text{ShelveLocMedium} + \beta_8 \cdot \text{Age} + \beta_9 \cdot \text{Education} + \beta_{10} \cdot \text{UrbanYes} + \beta_{11} \cdot \text{USYes} + \epsilon$$

Substituting the coefficients:

$$Y = 5.6606 + 0.0928 \cdot \text{CompPrice} + 0.0158 \cdot \text{Income} + 0.1231 \cdot \text{Advertising} + 0.0002 \cdot \text{Population} - 0.0954 \cdot \text{Price}$$

$$+4.8502 \cdot \text{ShelveLocGood} + 1.9567 \cdot \text{ShelveLocMedium} - 0.0460 \cdot \text{Age} - 0.0211 \cdot \text{Education} + 0.1229 \cdot \text{UrbanYes} \\ - 0.1841 \cdot \text{USYes} + \epsilon$$

Handling qualitative variables:

1. Shelve Location (ShelveLoc):

- ShelveLocGood = 1 if the shelving location is "Good," and 0 otherwise.
- ShelveLocMedium = 1 if the shelving location is "Medium," and 0 otherwise.
- "Bad" shelving location is the reference category (ShelveLocGood = 0 and ShelveLocMedium = 0).

2. Urban (Urban):

- UrbanYes = 1 if the location is urban, and 0 otherwise.
- "No" is the reference category (UrbanYes = 0).

3. US (US):

- USYes = 1 if the location is in the US, and 0 otherwise.
- "No" is the reference category (USYes = 0).

Model Interpretation:

- When all quantitative variables are set to 0 and all categorical variables are at their reference levels (ShelveLoc = "Bad", Urban = "No", US = "No") the predicted value of Y is 5.606
- Each coefficient represents the change in Y when the corresponding variable increases by one unit (for quantitative variables) or moves to a specific category (for qualitative variables), holding all else constant.

(d)

Coefficients Table

The table below shows the estimated coefficients, standard errors, t-values, and p-values for each predictor:

Predictor	Estimate	Std. Error	t-value	p-value	Reject H_0 ?
(Intercept)	5.6606	0.6034	9.380	< 2e-16	Yes
CompPrice	0.0928	0.0041	22.378	< 2e-16	Yes
Income	0.0158	0.0018	8.565	2.58e-16	Yes
Advertising	0.1231	0.0111	11.066	< 2e-16	Yes
Population	0.0002	0.0004	0.561	0.575	No
Price	-0.0954	0.0027	-35.700	< 2e-16	Yes
ShelveLocGood	4.8502	0.1531	31.678	< 2e-16	Yes
ShelveLocMedium	1.9567	0.1261	15.516	< 2e-16	Yes
Age	-0.0460	0.0032	-14.472	< 2e-16	Yes
Education	-0.0211	0.0197	-1.070	0.285	No
UrbanYes	0.1229	0.1130	1.088	0.277	No
USYes	-0.1841	0.1498	-1.229	0.220	No

Decision Rule

At a significance level of $\alpha = 0.05$, we reject $H_0 : \beta_j = 0$ for predictors with p-value < 0.05 .

Results

Based on the p-values:

- **Reject H_0 :**
 - CompPrice
 - Income
 - Advertising
 - Price
 - ShelfLocGood
 - ShelfLocMedium
 - Age
- **Fail to Reject H_0 :**
 - Population
 - Education
 - UrbanYes
 - USYes

Conclusion

The predictors CompPrice, Income, Advertising, Price, ShelfLocGood, ShelfLocMedium, and Age have a statistically significant relationship with the response variable Sales . The remaining predictors do not show significant effects at $\alpha = 0.05$.

(e)

Significant Predictors

From the previous analysis, the following predictors showed evidence of association with the outcome ($p < 0.05$):

- CompPrice
- Income
- Advertising
- Price
- ShelfLocGood
- ShelfLocMedium
- Age

R Code for Smaller Model

```
# Load necessary library (if needed)
# library(datasets) # or appropriate library for the data
# Fit the smaller model
lm.smaller <- lm(Sales ~ CompPrice + Income + Advertising + Price +
  ShelfLoc + ShelfLoc + Age, data = carseat)

# Summary of the smaller model
summary(lm.smaller)
```

Call:

```
lm(formula = Sales ~ CompPrice + Income + Advertising + Price +
  ShelfLoc + ShelfLoc + Age, data = carseat)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.7728	-0.6954	0.0282	0.6732	3.3292

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.475226	0.505005	10.84	<2e-16 ***
CompPrice	0.092571	0.004123	22.45	<2e-16 ***
Income	0.015785	0.001838	8.59	<2e-16 ***
Advertising	0.115903	0.007724	15.01	<2e-16 ***
Price	-0.095319	0.002670	-35.70	<2e-16 ***
ShelveLocGood	4.835675	0.152499	31.71	<2e-16 ***
ShelveLocMedium	1.951993	0.125375	15.57	<2e-16 ***
Age	-0.046128	0.003177	-14.52	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.019 on 392 degrees of freedom

Multiple R-squared: 0.872, Adjusted R-squared: 0.8697

F-statistic: 381.4 on 7 and 392 DF, p-value: < 2.2e-16

(f)

Comparing Model Fits: lm.reg1 (Model a) and lm.smaller (Model e)

Model (a): lm.reg1

- **Formula:** Sales ~ Price + Urban + US
- **Key Metrics:**
 - Residual Standard Error (RSE): **2.472**
 - Multiple R^2 : **0.2393**
 - Adjusted R^2 : **0.2335**
 - F-statistic: **41.52** ($p < 2.2 \times 10^{-16}$)

Model (e): lm.smaller

- **Formula:** Sales ~ CompPrice + Income + Advertising + Price + ShelveLocGood + ShelveLocMedium + Age
- **Key Metrics:**
 - Residual Standard Error (RSE): **1.019**
 - Multiple R^2 : **0.872**
 - Adjusted R^2 : **0.8697**
 - F-statistic: **381.4** ($p < 2.2 \times 10^{-16}$)

Comparison Table

Metric	Model (a): lm.reg1	Model (e): lm.smaller
Residual Standard Error (RSE)	2.472	1.019
Multiple R^2	0.2393	0.872
Adjusted R^2	0.2335	0.8697
F-statistic	41.52	381.4

Interpretation

Model (a):

- Only 23.93% of the variation in Sales is explained by the predictors Price, Urban, and US.
- The Residual Standard Error (RSE) is 2.472, indicating a high level of unexplained variability.
- The R^2 and Adjusted R^2 values are relatively low, showing that the model does not fit the data well.

Model (e):

- This model explains **87.2% of the variation in Sales**, a substantial improvement over Model (a).
- The Residual Standard Error (RSE) is significantly lower at **1.019**, showing that the model captures much more of the variability in Sales.
- The R^2 and Adjusted R^2 values indicate a strong fit, and the F-statistic confirms that the model is highly significant.

Conclusion

Model (e) fits the data far better than Model (a). It has a much higher R^2 , lower Residual Standard Error, and a higher F-statistic. This suggests that Model (e) is more effective at explaining the variation in Sales and should be preferred for inference or prediction.

Question 3

Load dataset(Credit.csv) and EDA

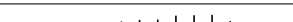
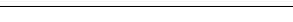
```
credit <- read.csv("Credit.csv")
head(credit)
```

```
  X Income Limit Rating Cards Age Education Own Student Married Region Balance
1 1 14.891 3606 283 2 34 11 No No Yes South 333
2 2 106.025 6645 483 3 82 15 Yes Yes Yes West 903
3 3 104.593 7075 514 4 71 11 No No No West 580
4 4 148.924 9504 681 3 36 11 Yes No No West 964
5 5 55.882 4897 357 2 68 16 No No Yes South 331
6 6 80.180 8047 569 4 77 10 No No No South 1151
```

```
str(credit)
```

```
'data.frame': 400 obs. of 12 variables:
 $ X      : int 1 2 3 4 5 6 7 8 9 10 ...
 $ Income : num 14.9 106 104.6 148.9 55.9 ...
 $ Limit  : int 3606 6645 7075 9504 4897 8047 3388 7114 3300 6819 ...
 $ Rating : int 283 483 514 681 357 569 259 512 266 491 ...
 $ Cards  : int 2 3 4 3 2 4 2 2 5 3 ...
 $ Age    : int 34 82 71 36 68 77 37 87 66 41 ...
 $ Education: int 11 15 11 11 16 10 12 9 13 19 ...
 $ Own    : chr "No" "Yes" "No" "Yes" ...
 $ Student : chr "No" "Yes" "No" "No" ...
 $ Married : chr "Yes" "Yes" "No" "No" ...
 $ Region : chr "South" "West" "West" "West" ...
 $ Balance : int 333 903 580 964 331 1151 203 872 279 1350 ...
```

```
library(Hmisc)
latex(describe(credit), file = "", caption.placement = "top")
```

credit													
12 Variables 400 Observations													
X													
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95	
400	0	400	1	200.5	133.7	20.95	40.90	100.75	200.50	300.25	360.10	380.05	
lowest : 1 2 3 4 5, highest: 396 397 398 399 400													
Income													
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95	
400	0	399	1	45.22	35.34	12.07	14.58	21.01	33.12	57.47	92.45	124.35	
lowest : 10.354 10.363 10.403 10.503 10.588 , highest: 163.329 180.379 180.682 182.728 186.634													
Limit													
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95	
400	0	387	1	4736	2546	1483	1919	3088	4622	5873	7660	9162	
lowest : 855 886 905 906 1134, highest: 11589 11966 12066 13414 13913													
Rating													
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95	
400	0	283	1	354.9	170.5	138.0	167.0	247.2	344.0	437.2	549.5	642.7	
lowest : 93 103 112 115 117, highest: 817 828 832 949 982													
Cards													
n	missing	distinct	Info	Mean	Gmd								
400	0	9	0.946	2.958	1.486								
Value 1 2 3 4 5 6 7 8 9													
Frequency 51 115 111 72 34 11 4 1 1													
Proportion 0.128 0.288 0.278 0.180 0.085 0.028 0.010 0.002 0.002													
For the frequency table, variable is rounded to the nearest 0													
Age													
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95	
400	0	68	1	55.67	19.9	29.00	32.00	41.75	56.00	70.00	79.10	82.00	
lowest : 23 24 25 26 27, highest: 86 87 89 91 98													
Education													
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95	
400	0	16	0.991	13.45	3.551	8	9	11	14	16	17	18	
Value 5 6 7 8 9 10 11 12 13 14 15 16 17 18													
Frequency 1 5 8 14 25 24 33 37 38 48 49 50 34 22													
Proportion 0.002 0.013 0.020 0.035 0.062 0.060 0.082 0.092 0.095 0.120 0.122 0.125 0.085 0.055													
Value 19 20													
Frequency 10 2													
Proportion 0.025 0.005													
For the frequency table, variable is rounded to the nearest 0													
Own													
n	missing	distinct											
400	0	2											
Value No Yes													
Frequency 193 207													
Proportion 0.482 0.517													
Student													
n	missing	distinct											
400	0	2											
Value No Yes													
Frequency 360 40													
Proportion 0.9 0.1													

Married

```
n    missing    distinct
400      0         2

Value    No    Yes
Frequency 155  245
Proportion 0.388 0.613
```

Region

```
n    missing    distinct
400      0         3

Value    East South West
Frequency  99  199  102
Proportion 0.248 0.498 0.255
```

Balance

```
n missing distinct Info Mean  Gmd .05 .10 .25 .50
400      0      284 0.989 520  517  0.00  0.00 68.75 459.50
75  90  95
863.00 1151.40 1355.30
lowest:  0  5  8 15 16, highest: 1677 1687 1779 1809 1999
```

M1

```
attach(credit)
```

```
# M1
lm.m1 <- lm(Balance~ Income)
summary(lm.m1)
```

Call:

```
lm(formula = Balance ~ Income)
```

Residuals:

```
Min      1Q  Median      3Q      Max
-803.64 -348.99 -54.42  331.75 1100.25
```

Coefficients:

```
Estimate Std. Error t value Pr(>|t|)
(Intercept) 246.5148   33.1993   7.425 6.9e-13 ***
Income      6.0484    0.5794  10.440 < 2e-16 ***
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 407.9 on 398 degrees of freedom

Multiple R-squared: 0.215, Adjusted R-squared: 0.213

F-statistic: 109 on 1 and 398 DF, p-value: < 2.2e-16

M3

```
# M3
lm.m3 <- lm(Balance~ Income + Student)
summary(lm.m3)
```

Call:

```
lm(formula = Balance ~ Income + Student)
```

Residuals:

Min	1Q	Median	3Q	Max
-762.37	-331.38	-45.04	323.60	818.28

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	211.1430	32.4572	6.505	2.34e-10 ***
Income	5.9843	0.5566	10.751	< 2e-16 ***
StudentYes	382.6705	65.3108	5.859	9.78e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 391.8 on 397 degrees of freedom
Multiple R-squared: 0.2775, Adjusted R-squared: 0.2738
F-statistic: 76.22 on 2 and 397 DF, p-value: < 2.2e-16

M4

```
# M4
```

```
lm.m4 <- lm(Balance~ Income + Income*Student)  
summary(lm.m4)
```

Call:

```
lm(formula = Balance ~ Income + Income * Student)
```

Residuals:

Min	1Q	Median	3Q	Max
-773.39	-325.70	-41.13	321.65	814.04

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	200.6232	33.6984	5.953	5.79e-09 ***
Income	6.2182	0.5921	10.502	< 2e-16 ***
StudentYes	476.6758	104.3512	4.568	6.59e-06 ***
Income:StudentYes	-1.9992	1.7313	-1.155	0.249

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 391.6 on 396 degrees of freedom
Multiple R-squared: 0.2799, Adjusted R-squared: 0.2744
F-statistic: 51.3 on 3 and 396 DF, p-value: < 2.2e-16

M5

```
# M5
```

```
lm.m5 <- lm(Balance~ Income + Student + Income*Student)  
summary(lm.m5)
```

Call:

```
lm(formula = Balance ~ Income + Student + Income * Student)
```

Residuals:

Min	1Q	Median	3Q	Max
-773.39	-325.70	-41.13	321.65	814.04

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	200.6232	33.6984	5.953	5.79e-09 ***
Income	6.2182	0.5921	10.502	< 2e-16 ***
StudentYes	476.6758	104.3512	4.568	6.59e-06 ***
Income:StudentYes	-1.9992	1.7313	-1.155	0.249

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 391.6 on 396 degrees of freedom

Multiple R-squared: 0.2799, Adjusted R-squared: 0.2744

F-statistic: 51.3 on 3 and 396 DF, p-value: < 2.2e-16

Model Summaries:

1. **M1 (Balance ~ Income):**

- Adjusted R^2 : 0.213
- Residual standard error: 407.9
- Interpretation: A simple linear regression where Income significantly predicts Balance ($p < 0.001$).

2. **M3 (Balance ~ Income + Student):**

- Adjusted R^2 : 0.2738
- Residual standard error: 391.8
- Interpretation: Adding the "Student" variable improves the model fit. Both Income ($p < 0.001$) and the Student indicator ($p < 0.001$) are statistically significant.

3. **M4 (Balance ~ Income + Income * Student):**

- Adjusted R^2 : 0.2744
- Residual standard error: 391.6
- Interpretation: Includes an interaction term between Income and Student. However, the interaction term ($p = 0.249$) is not statistically significant, indicating that it does not contribute much to the model.

4. **M5 (Balance ~ Income + Student + Income * Student):**

- Adjusted R^2 : 0.2744
- Residual standard error: 391.6
- Interpretation: Similar to M4. The interaction term remains insignificant.

Conclusion:

- **Best Model: M3 (Balance ~ Income + Student)**

- **Reasons:**

1. It has a higher adjusted R^2 (0.2738) than M1, indicating improved explanatory power.
2. Both predictors (Income and Student) are statistically significant.
3. The model avoids the unnecessary complexity of the insignificant interaction term present in M4 and M5, maintaining parsimony.

Model M3 strikes a balance between fit and simplicity. Use M3 for predicting "Balance" as it effectively captures the relationship without overcomplicating the model.